

ADVANCED

Stage 6

Calculus (Adv), C1 Introduction to Differentiation (Adv)

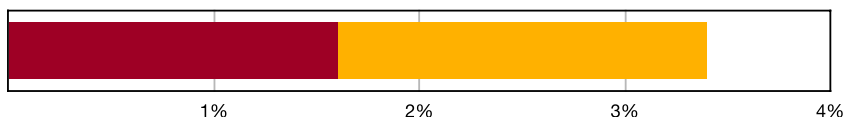
Rates of Change (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 48 minutes (based on allocation of 1.5 minutes per mark)



C1 Introduction to Differentiation C2 Differential Calculus



***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- C1 Introduction to Differentiation* is Y11 content that has contributed an average of 1.6% per exam since the new syllabus was introduced in 2020.
- This topic has been split into three sub-topics for analysis purposes: 1-Standard Differentiation (0.8%), 2-Tangents (0.8%), and 3-Rates of Change.
- This analysis looks at *Rates of Change*.

HSC ANALYSIS - What to expect and common pitfalls

- Rates of Change*, within *C1 Introduction to Differentiation*, only includes examples that require standard differentiation. This excludes log, exponential and trig underlyings and any questions that require integration - questions involving these aspects of *Rates of Change* are covered later in year 12.
- Rates of Change* questions that require standard differentiation have not been asked in new syllabus exams to date. Before this however, they were examined four times in the period 2013-2019, primarily as multiple choice questions but including a significant 5 mark allocation in 2018.
- This sub-topic provides important gateway content for the challenging year 12 work in this area.

Questions

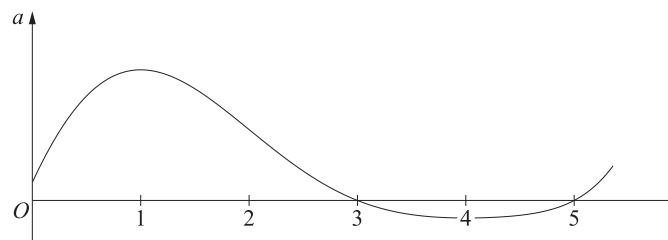
1. Calculus, 2ADV C1 EQ-Bank 2 MC

The displacement of a particle is given by $x = 3t^3 - 6t^2 - 15$. The acceleration is zero at:

- $t = \frac{2}{3}$
- $t = \frac{4}{3}$
- $t = \frac{5}{2}$
- never

2. Calculus, 2ADV C1 2019 HSC 8 MC

A particle is moving along a straight line. The graph shows the acceleration of the particle.

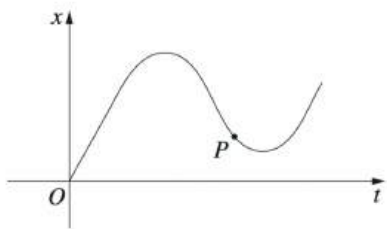


For what value of t is the velocity v a maximum?

- 1
- 2
- 3
- 5

3. Calculus, 2ADV C1 2014 HSC 9 MC

The graph shows the displacement x of a particle moving along a straight line as a function of time t .



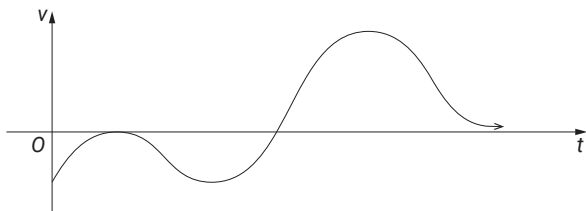
Which statement describes the motion of the particle at the point P ?

- A. The velocity is negative and the acceleration is positive.
- B. The velocity is negative and the acceleration is negative.
- C. The velocity is positive and the acceleration is positive.
- D. The velocity is positive and the acceleration is negative.

4. Calculus, 2ADV C1 2017 HSC 10 MC

A particle is moving along a straight line.

The graph shows the velocity, v , of the particle for time $t \geq 0$.

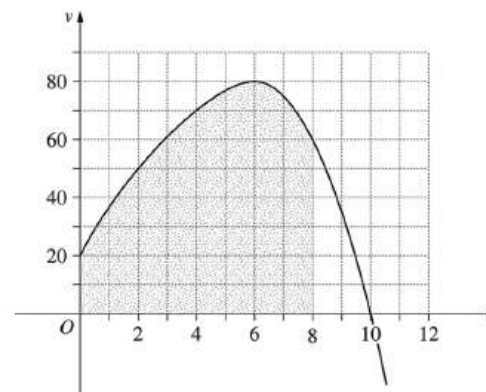


How many times does the particle change direction?

- A. 1
- B. 2
- C. 3
- D. 4

5. Calculus, 2ADV C1 2008 HSC 6b

The graph shows the velocity of a particle, v metres per second, as a function of time, t seconds.



- i. What is the initial velocity of the particle? **(1 mark)**
- ii. When is the velocity of the particle equal to zero? **(1 mark)**
- iii. When is the acceleration of the particle equal to zero? **(1 mark)**

6. Calculus, 2ADV C1 2018 HSC 12d

The displacement of a particle moving along the x -axis is given by

$$x = \frac{t^3}{3} - 2t^2 + 3t,$$

where x is the displacement from the origin in metres and t is the time in seconds, for $t \geq 0$.

- i. What is the initial velocity of the particle? **(1 mark)**
- ii. At which times is the particle stationary? **(2 marks)**
- iii. Find the position of the particle when the acceleration is zero. **(2 marks)**

7. Calculus, 2ADV C1 EQ-Bank 5

A magpie plague hit Raymond Terrace this year but was eventually brought under control. A bird researcher estimated that the magpie population M , in hundreds, t months after 1st January, was given by $M = 7 + 20t - 3t^2$

- a. Find the magpie population on 1st March. **(1 mark)**
- b. At what rate was the population changing at this time? **(1 mark)**
- c. In what month does the magpie population start to decrease? **(2 marks)**

8. Calculus, 2ADV C1 SM-Bank 3

The displacement x metres from the origin at time t seconds of a particle travelling in a straight line is given by

$$x = 2t^3 - t^2 - 3t + 11 \quad \text{when } t \geq 0$$

- Calculate the velocity when $t = 2$. (1 mark)
- When is the particle stationary? (2 marks)

9. Calculus, 2ADV C1 SM-Bank 11

A particle is moving along the x -axis. Its velocity v at time t is given by

$$v = \sqrt{20t - 2t^2} \text{ metres per second}$$

Find the acceleration of the particle when $t = 4$.

Express your answer as an exact value in its simplest form. (3 marks)

10. Calculus, 2ADV C1 2014 HSC 13c

The displacement of a particle moving along the x -axis is given by

$$x = t - \frac{1}{1+t},$$

where x is the displacement from the origin in metres, t is the time in seconds, and $t \geq 0$.

- Show that the acceleration of the particle is always negative. (2 marks)
- What value does the velocity approach as t increases indefinitely? (1 mark)

11. Calculus, 2ADV C1 2006 HSC 8a

A particle is moving in a straight line. Its displacement, x metres, from the origin, O , at time t seconds, where $t \geq 0$, is given by $x = 1 - \frac{7}{t+4}$.

- Find the initial displacement of the particle. (1 mark)
- Find the velocity of the particle as it passes through the origin. (3 marks)
- Show that the acceleration of the particle is always negative. (1 mark)
- Sketch the graph of the displacement of the particle as a function of time. (2 marks)

Worked Solutions

1. Calculus, 2ADV C1 EQ-Bank 2 MC

$$x = 3t^3 - 6t^2 - 15$$

$$v = 9t^2 - 12t$$

$$a = 18t - 12$$

Find t when $a = 0$:

$$18t - 12 = 0 \Rightarrow t = \frac{2}{3}$$

$\Rightarrow A$

2. Calculus, 2ADV C1 2019 HSC 8 MC

Velocity increases when $a > 0$.

$\therefore v_{\max}$ occurs when $t = 3$.

$\Rightarrow C$

3. Calculus, 2ADV C1 2014 HSC 9 MC

At P , the particle is moving back towards O after hitting a max (positive) displacement

$\therefore$ Velocity is negative.

Its displacement hits a minimum just after P and increases again.

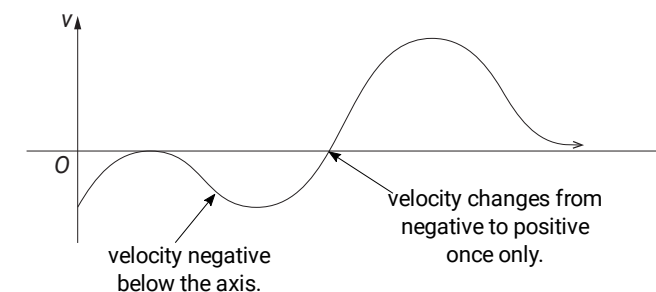
$\therefore$ Acceleration is working against (negative) velocity and must be positive.

$\Rightarrow A$

4. Calculus, 2ADV C1 2017 HSC 10 MC

$\Rightarrow A$

◆◆◆ Mean mark 33%.



5. Calculus, 2ADV C1 2008 HSC 6b

- i. Find v when $t = 0$
- $$v = 20 \text{ m/s}$$
- ii. Particle comes to rest at $t = 10$ seconds (from graph)
- iii. Acceleration is zero when $t = 6$ seconds (from graph)

6. Calculus, 2ADV C1 2018 HSC 12d

i. $x = \frac{t^3}{3} - 2t^2 + 3t$

$$v = \frac{dx}{dt} = t^2 - 4t + 3$$

Find v when $t = 0$:

$$v = 0 - 0 + 3$$
$$= 3 \text{ ms}^{-1}$$

- ii. Particle is stationary when $v = 0$

$$t^2 - 4t + 3 = 0$$
$$(t - 3)(t - 1) = 0$$
$$t = 1 \text{ or } 3 \text{ seconds}$$

iii. $a = \frac{dv}{dt} = 2t - 4$

Find t when $a = 0$

$$2t - 4 = 0$$
$$t = 2$$
$$x(2) = \frac{2^3}{3} - 2(2^2) + 3(2)$$
$$= \frac{8}{3} - 8 + 6$$
$$= \frac{2}{3}$$

7. Calculus, 2ADV C1 EQ-Bank 5

a. Find M when $t = 2$:

$$M = 7 + 20 \times 2 - 3 \times 2^2 = 35$$
$$\therefore \text{Population} = 35 \times 100 = 3500$$

b. $M = 7 + 20t - 3t^2$

$$\frac{dM}{dt} = 20 - 6t$$

Find $\frac{dM}{dt}$ when $t = 2$:

$$\frac{dM}{dt} = 20 - 6 \times 2 = 8$$
$$\therefore M \text{ is increasing at } 800 \text{ per month}$$

c. Find t when $\frac{dM}{dt} = 0$:

$$\frac{dM}{dt} = 20 - 6t = 0 \Rightarrow t = 3 \frac{1}{3}$$
$$\frac{dM}{dt} < 0 \text{ when } t > 3 \frac{1}{3}$$
$$\therefore \text{Population starts to decrease in May.}$$

8. Calculus, 2ADV C1 SM-Bank 3

i. $x = 2t^3 - t^2 - 3t + 11$

$$v = \frac{dx}{dt} = 6t^2 - 2t - 3$$

When $t = 2$,

$$v = 6 \times 2^2 - 2 \cdot 2 - 3$$
$$= 17 \text{ ms}^{-1}$$

- ii. Particle is stationary when $v = 0$

$$6t^2 - 2t - 3 = 0$$
$$\therefore t = \frac{2 \pm \sqrt{(-2)^2 - 4 \cdot 6 \cdot (-3)}}{12}$$
$$= \frac{2 \pm \sqrt{76}}{12}$$
$$= \frac{1 \pm \sqrt{19}}{6}$$
$$= \frac{1 + \sqrt{19}}{6} \quad (t \geq 0)$$

9. Calculus, 2ADV C1 SM-Bank 11

$$v = \sqrt{20t - 2t^2}$$

$$\begin{aligned}\alpha &= \frac{dv}{dt} \\ &= \frac{1}{2} \cdot (20t - 2t^2)^{-\frac{1}{2}} \cdot (20 - 4t)\end{aligned}$$

When $t = 4$,

$$\begin{aligned}\alpha &= \frac{1}{2} (20 \cdot 4 - 2 \cdot 4^2)^{-\frac{1}{2}} (20 - 16) \\ &= \frac{2}{\sqrt{48}} \\ &= \frac{2}{4\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{\sqrt{3}}{6} \text{ ms}^{-2}\end{aligned}$$

10. Calculus, 2ADV C1 2014 HSC 13c

i.
$$\begin{aligned}x &= t - \frac{1}{1+t} \\ &= t - (1+t)^{-1}\end{aligned}$$

$$\begin{aligned}\dot{x} &= 1 - (-1)(1+t)^{-2} \\ &= 1 + \frac{1}{(1+t)^2}\end{aligned}$$

$$\begin{aligned}\ddot{x} &= -2(1+t)^{-3} \\ &= -\frac{2}{(1+t)^3}\end{aligned}$$

Since $t \geq 0$,

$$\Rightarrow -\frac{2}{(1+t)^3} < 0$$

$\therefore$ Acceleration is always negative.

ii. Velocity $(\dot{x}) = 1 + \frac{1}{(1+t)^2}$

$$\text{As } t \rightarrow \infty, \frac{1}{(1+t)^2} \rightarrow 0$$

$$\therefore \text{As } t \rightarrow \infty, \dot{x} \rightarrow 1$$

11. Calculus, 2ADV C1 2006 HSC 8a

i. $x = 1 - \frac{7}{t+4}$

When $t = 0$,

$$\begin{aligned} x &= 1 - \frac{7}{4} \\ &= -\frac{3}{4} \text{ m} \end{aligned}$$

$\therefore$ Initial displacement is $\frac{3}{4}$ metres to the left of the origin.

ii. $x = 1 - \frac{7}{t+4} = 1 - 7(t+4)^{-1}$

$$\begin{aligned} \dot{x} &= (-1) - 7(t+4)^{-2} \times \frac{d}{dt}(t+4) \\ &= 7(t+4)^{-2} \times 1 \\ &= \frac{7}{(t+4)^2} \end{aligned}$$

Find t when $x = 0$

$$\begin{aligned} 0 &= 1 - \frac{7}{t+4} \\ \frac{7}{t+4} &= 1 \\ 7 &= (t+4) \\ t &= 3 \end{aligned}$$

When $t = 3$

$$\begin{aligned} \dot{x} &= \frac{7}{(3+4)^2} \\ &= \frac{1}{7} \text{ ms}^{-1} \end{aligned}$$

$\therefore$ The velocity of the particle as it passes through the origin is $\frac{1}{7} \text{ ms}^{-1}$.

iii. $\dot{x} = 7(t+4)^{-2}$

$$\ddot{x} = \frac{d\dot{x}}{dt} = -14(t+4)^{-3}$$

Given $t \geq 0$

$$\Rightarrow (t+4)^{-3} \geq 0$$

$$\Rightarrow -14(t+4)^{-3} \leq 0$$

$\therefore \ddot{x}$ is always negative.

